i. Bruteforce

It will try all possible combinations for the passwords. Time taken to crack the password increases exponentially with time. Bruteforce attacks work best if the WPS pin is still enabled in the router.

Here's what Wikipedia says about WPS:

"Created by the Wi-Fi Alliance and introduced in 2006, the goal of the protocol is to allow home users who know little of wireless security and may be intimidated by the available security options to set up Wi-Fi Protected Access, as well as making it easy to add new devices to an existing network without entering long passphrases. Before the standard, several competing solutions were developed by different vendors to address the same need. A major security flaw was revealed in December 2011 that affects wireless routers with the WPS feature, which most recent models have enabled by default. The flaw allows a remote attacker to recover the WPS PIN in a few hours with a brute-force attack and, with the WPS PIN, the network's WPA/WPA2 pre-shared key. Users have been urged to turn off the WPS feature, although this may not be possible on some router models."

It is an 8 digit verification number to access the network. Validation of keys takes time, almost one second per key. So theoretically it would take 10^8 seconds, which is close to three years, but there is a sleek workaround. The 8th digit is a checksum digit. So we only have to check the first seven. The pin number goes in two halves for verification. The first half would take only 104 guesses and the second half would take mere 103.

So a total guess needed would be 11,000, which would take close to three hours.

ii. Wordlist/Dictionary

There is a predefined list of guesses (a word list or dictionary of passwords). These word list files contain commonly used passwords, combinations of passwords, misspelling of words, a combination of words and numbers etc. The good dictionaries may be a few GBs in size and can be used to hack most common networks. This method won't work if the networks password isn't on the word list. With a dictionary of decent size, the attack takes a reasonable amount of time.

iii. Rainbow table

The validation process against the 4-way handshake involves hashing of the password and comparing it with the hash in a handshake. Hashing